

ALFAAJ67457

## Gold nanoparticles, 250nm, supplied in 0.1mM PBS

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** Gold nanoparticles, 250nm, supplied in 0.1mM PBS  
Gold nanoparticles, 250nm, supplied in 0.1mM PBS

**Cat No. :** J67457

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

|                                                    |                                               |                         |
|----------------------------------------------------|-----------------------------------------------|-------------------------|
| <b>Physical State</b><br>Liquid                    | <b>Appearance</b><br>No information available | <b>Odor</b><br>Odorless |
| <b>Emergency Overview</b><br>Sensitivity to light. |                                               |                         |

#### Classification of the substance or mixture

Based on available data, the classification criteria are not met

#### Label Elements

None required

#### **Physical and Chemical Hazards**

None identified.

#### **Health Hazards**

The product contains no substances which at their given concentration are considered to be hazardous to health.

#### **Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

This product does not contain any known or suspected endocrine disruptors.

**SAFETY DATA SHEET**

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**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                      | CAS No    | Weight % |
|--------------------------------|-----------|----------|
| Water                          | 7732-18-5 | 99.9804  |
| Gold                           | 7440-57-5 | 0.01     |
| Sodium chloride                | 7647-14-5 | 0.008    |
| Sodium phosphate dibasic       | 7558-79-4 | 0.0011   |
| Dihydrogen potassium phosphate | 7778-77-0 | 0.0003   |
| Potassium chloride             | 7447-40-7 | 0.0002   |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

No special precautions required.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Not combustible.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

None reasonably foreseeable.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

# SAFETY DATA SHEET

## Gold nanoparticles, 250nm, supplied in 0.1mM PBS

### Methods for Containment and Clean Up

Sweep up and shovel into suitable containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7. HANDLING AND STORAGE

### Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

### Storage

Keep refrigerated.

### Specific Use(s)

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Monitoring methods

MDHS 91 Metals and metalloids in workplace air by X-ray fluorescence spectrometry MDHS 99 Metals in air by ICP-AES

### Exposure Controls

### Engineering Measures

None under normal use conditions. .

### Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Nitrile rubber |                                   |                 |             |                       |
| Neoprene       |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particle filter

## Small scale/Laboratory use

Maintain adequate ventilation

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## Environmental exposure controls

No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

## Appearance

## Physical State

Liquid

## Odor

Odorless

## Odor Threshold

No data available

## pH

5 - 7

## Melting Point/Range

No data available

## Softening Point

No data available

## Boiling Point/Range

No information available

## Flash Point

No information available

Method - No information available

## Evaporation Rate

No data available

## Flammability (solid,gas)

Not applicable

Liquid

## Explosion Limits

No data available

## Vapor Pressure

23 hPa @ 20 °C

## Vapor Density

No data available

(Air = 1.0)

## Specific Gravity / Density

No data available

## Bulk Density

Not applicable

Liquid

## Water Solubility

Miscible

## Solubility in other solvents

No information available

## Partition Coefficient (n-octanol/water)

## Autoignition Temperature

No data available

## Decomposition Temperature

No data available

## Viscosity

No data available

## Explosive Properties

No information available

## Oxidizing Properties

No information available

## SECTION 10. STABILITY AND REACTIVITY

## Stability

Light sensitive.

## Hazardous Reactions

None under normal processing.

## Hazardous Polymerization

No information available.

## Conditions to Avoid

None known.

## Materials to avoid

No information available.

Hazardous Decomposition Products Hydrogen chloride. Oxides of phosphorus. Potassium oxides. Sodium oxides. Gold oxide.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

Toxicology data for the components

**SAFETY DATA SHEET****Gold nanoparticles, 250nm, supplied in 0.1mM PBS**

| Component                      | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation              |
|--------------------------------|---------------------------|-------------------------------|------------------------------|
| Water                          | -                         | -                             | -                            |
| Sodium chloride                | LD50 = 3 g/kg ( Rat )     | LD50 > 10000 mg/kg ( Rabbit ) | LC50 > 42 mg/L ( Rat ) 1 h   |
| Sodium phosphate dibasic       | LD50 = 17 g/kg ( Rat )    |                               |                              |
| Dihydrogen potassium phosphate | LD50 = 3200 mg/kg ( Rat ) | LD50 > 4640 mg/kg ( Rabbit )  | LC50 > 0.83 mg/L ( Rat ) 4 h |
| Potassium chloride             | LD50 = 2600 mg/kg ( Rat ) |                               |                              |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available

Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects,both acute and delayed No information available

**SECTION 12. ECOLOGICAL INFORMATION****Ecotoxicity effects**

May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component          | Freshwater Fish                                                                                    | Water Flea          | Freshwater Algae    | Microtox |
|--------------------|----------------------------------------------------------------------------------------------------|---------------------|---------------------|----------|
| Sodium chloride    | Pimephals prome:<br>LC50: 7650 mg/L/96h                                                            | EC50: 1000 mg/L/48h |                     |          |
| Potassium chloride | Lepomis macrochirus:<br>LC50: 1060 mg/L /96h<br>Pimephales promelas:<br>LC50: 750 - 1020 mg/L /96h | EC50: 825 mg/L/48h  | EC50: 2500 mg/L/72h |          |

**Persistence and Degradability****Persistence**

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary based on information available, May persist.

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|                                              |                                                                                                                                                           |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Degradation in sewage treatment plant</b> | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.                                           |
| <b>Bioaccumulative Potential</b>             | May have some potential to bioaccumulate                                                                                                                  |
| <b>Mobility in soil</b>                      | The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils |
| <b>Endocrine Disruptor Information</b>       | This product does not contain any known or suspected endocrine disruptors                                                                                 |
| <b>Persistent Organic Pollutant</b>          | This product does not contain any known or suspected substance                                                                                            |
| <b>Ozone Depletion Potential</b>             | This product does not contain any known or suspected substance                                                                                            |

## SECTION 13. DISPOSAL CONSIDERATIONS

|                                            |                                                                                                                                                                                                                             |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
| <b>Contaminated Packaging</b>              | Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.                                                                                                                  |
| <b>Other Information</b>                   | Waste codes should be assigned by the user based on the application for which the product was used.                                                                                                                         |

## SECTION 14. TRANSPORT INFORMATION

**Road and Rail Transport** Not Regulated

**IMDG/IMO** Not regulated

**IATA** Not regulated

**Special Precautions for User** No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                      | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|--------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Water                          | -                                                   | -                                       | X    | X     | 231-791-2 | X    | X   | X     | X    |      | X    | KE-35400 |
| Gold                           | -                                                   | -                                       | X    | X     | 231-165-9 | X    | X   | X     | X    |      | X    | KE-18083 |
| Sodium chloride                | -                                                   | -                                       | X    | X     | 231-598-3 | X    | X   | X     | X    | X    | X    | KE-31387 |
| Sodium phosphate dibasic       | -                                                   | -                                       | X    | X     | 231-448-7 | X    | X   | X     | X    | X    | X    | KE-12344 |
| Dihydrogen potassium phosphate | -                                                   | -                                       | X    | X     | 231-913-4 | X    | X   | X     | X    | X    | X    | KE-28622 |
| Potassium chloride             | -                                                   | -                                       | X    | X     | 231-211-8 | X    | X   | X     | X    | X    | X    | KE-29086 |

# SAFETY DATA SHEET

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### National Regulations

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Revision Date** 08-May-2024  
**Revision Summary** New emergency telephone response service provider.

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Physical hazards** On basis of test data  
**Health Hazards** Calculation method  
**Environmental hazards** Calculation method

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**